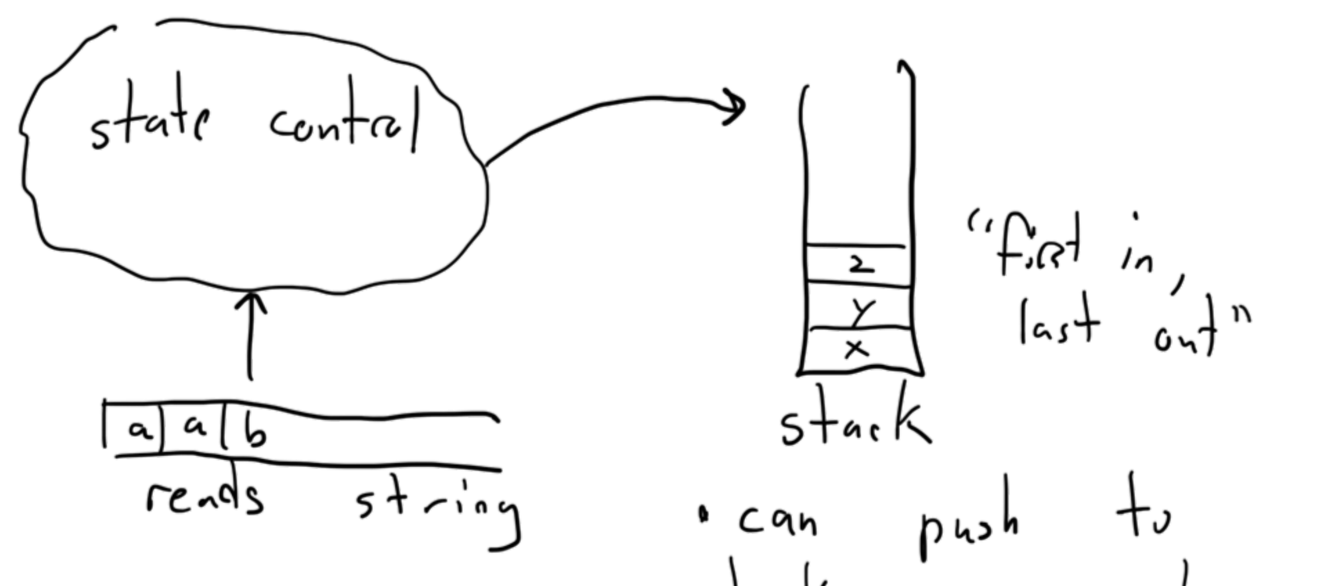
Pushdown Automata (PDA)

- will recognize CFL's, just like CFG's

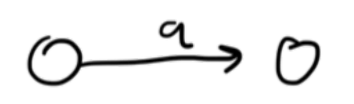


- can push to stack, or read+pop from top of stack

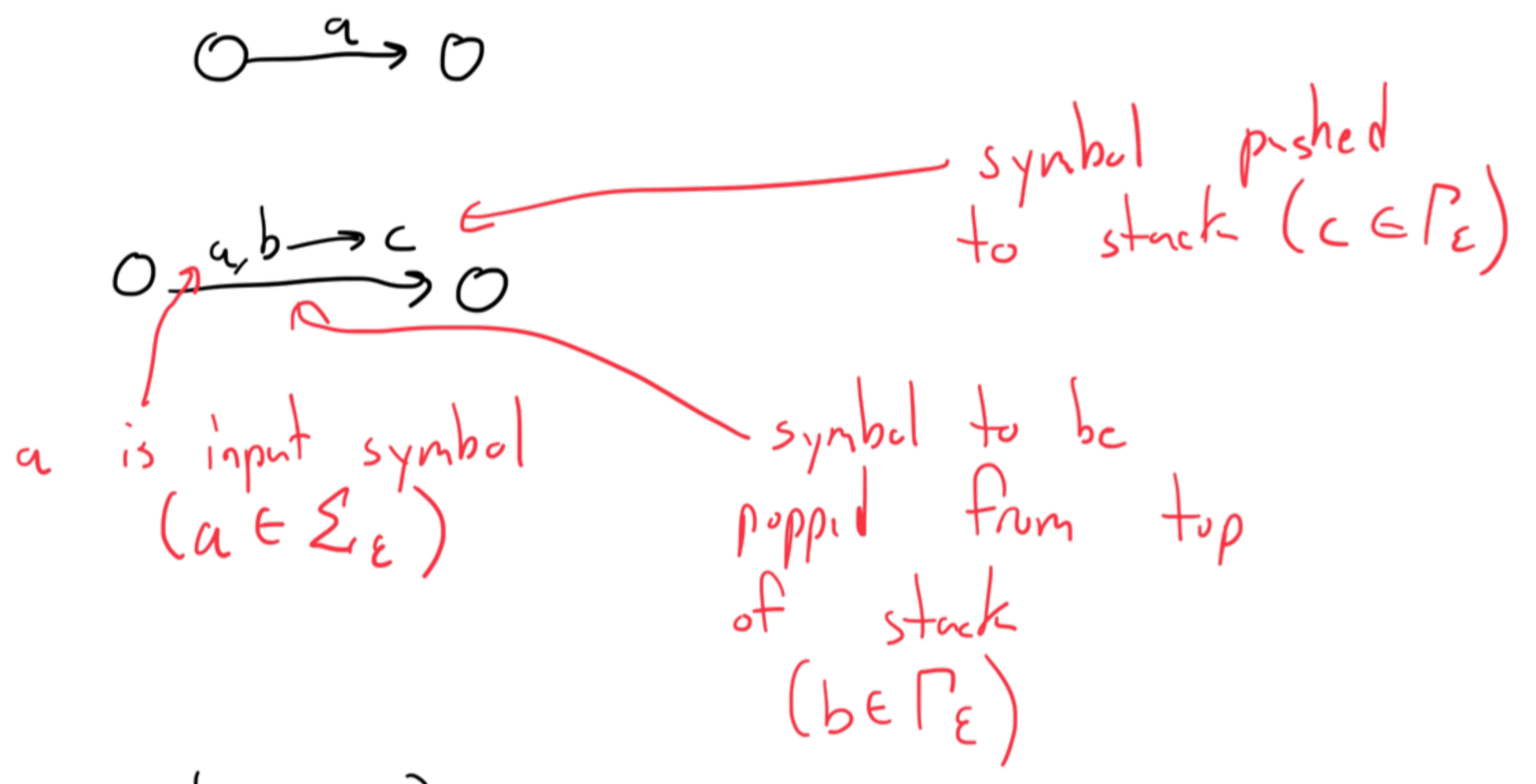
- A PDA is a 6-tuple $(Q, \Sigma, \Gamma, \delta, q_0, F)$ where:
  1. Q is the set of states
  2. Σ is the input alphabet
  3. Γ is the stack alphabet
  4. $\delta: Q \times \Sigma_\varepsilon \times \Gamma_\varepsilon \to P(Q \times \Gamma_\varepsilon)$ is the transition function
  5. $q_0 \in Q$ is the start state
  6. $F \subseteq Q$ is the set of accepting states.

- We assume PDA's are nondeterministic. NPDAs are strictly more powerful than DPDAs.

FSM



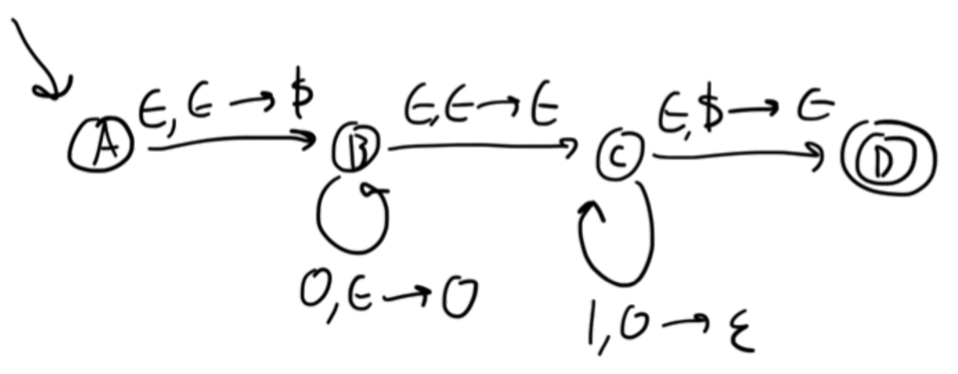
PDA



$\{0^n 1^n \mid n \geq 0\}$

$\Sigma = \{0, 1\}$ input

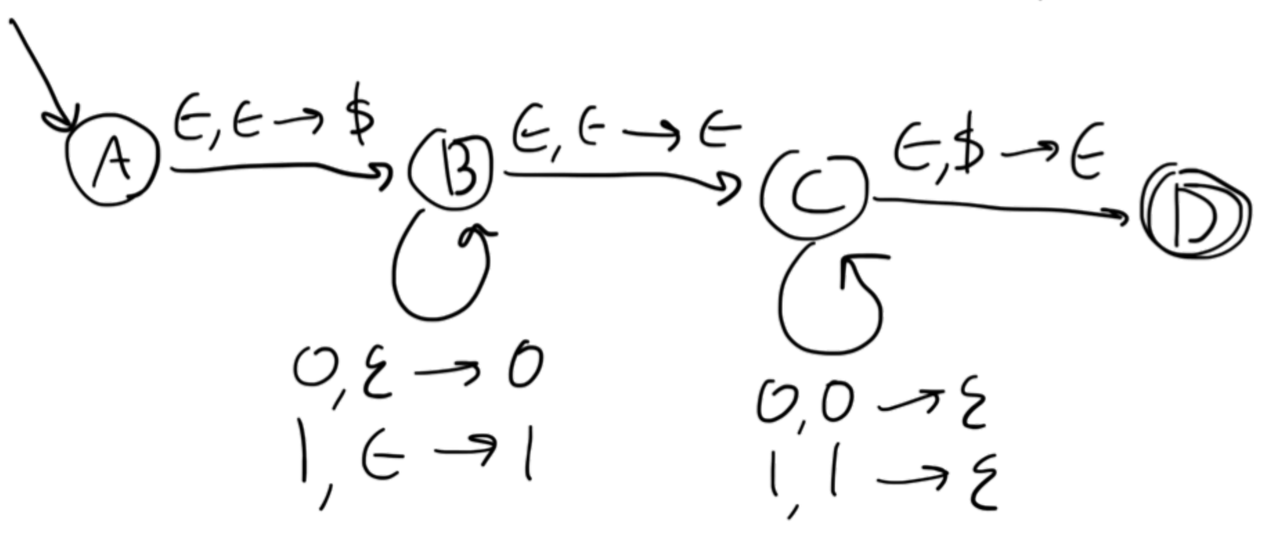
$\Gamma = \{\$, 0\}$ stack



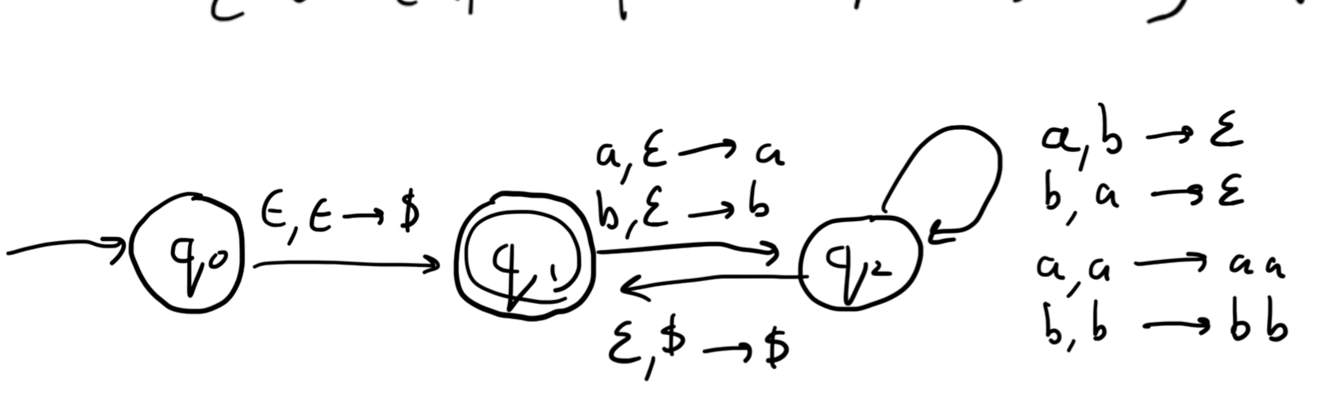
- machine can end with symbols on the stack (but this machine does not)

$L = \{w \in \{0,1\}^* \mid w$ is a palindrome and has an even # of symbols$\}$

$S \to 0S0 \mid 1S1 \mid \varepsilon$

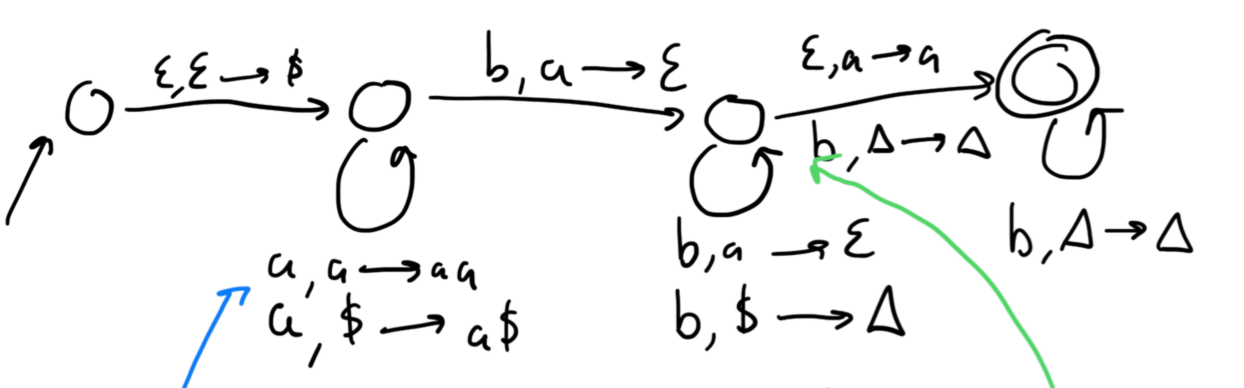


$L = \{w \in \{a,b\}^* \mid \#_a(w) = \#_b(w)\}$ ← # of a's is equal to the # of b's.



Try "abaabb"

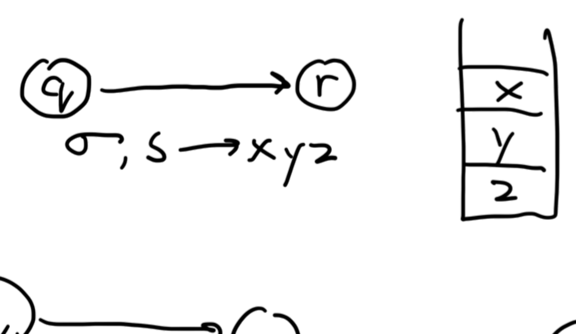
$C = \{a^m b^n : m \neq n$ and $n, m > 0\}$

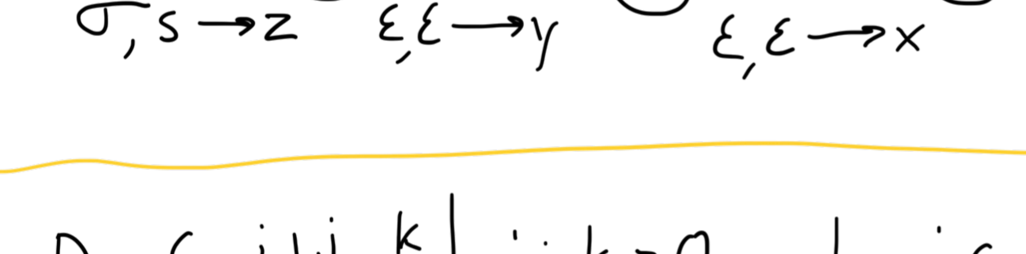


Try "aabbb" and "aabb" and "aab"

This input will get stuck at the 3rd node!

Shorthand





$D = \{a^i b^j c^k \mid i,j,k \geq 0$ and $i \in \{j, k\}\}$

$i = j$ or $i = k$